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ORIGINAL INVESTIGATION

Sleep Quality but Not Quantity Altered With a Change in Training Environment in Elite Australian Rules Football Players

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Purpose: To assess the effects of a change in training environment on the sleep characteristics of elite Australian Rules football (AF) players. ***Methods:*** In an observational crossover trial, 19 elite AF players had time in bed (TIB), total sleep time (TST), sleep efficiency (SE), and wake after sleep onset (WASO) assessed using wristwatch activity devices and subjective sleep diaries across 8-d home and camp periods. Repeated-measures ANOVA determined mean differences in sleep, training load (session rating of perceived exertion [RPE]), and environment. Pearson product–moment correlations, controlling for repeated observations on individuals, were used to assess the relationship between changes in sleep characteristics at home and camp. Cohen effect sizes (*d*) were calculated using individual means. ***Results:*** On camp TIB (+34 min) and WASO (+26 min) increased compared with home. However, TST was similar between home and camp, significantly reducing camp SE (–5.82%). Individually, there were strong negative correlations for TIB and WASO (r = -.75 and r = -.72, respectively) and a moderate negative correlation for SE (r = -.46) between home and relative changes on camp. Camp increased the relationship between individual s-RPE variation and TST variation compared with home (increased load r = -.367 vs .051, reduced load r = .319 vs –.033, camp vs home respectively). ***Conclusions:*** Camp compromised sleep quality due to significantly increased TIB without increased TST. Individually, AF players with higher home SE experienced greater reductions in SE on camp. Together, this emphasizes the importance of individualized interventions for elite team-sport athletes when traveling and/or changing environments.

Keywords: team sports, travel, athletes, ActiGraphy

Australian Rules football (AF) clubs commonly undertake preseason training camps to improve team cohesion and morale and to maximize training adaptations using specific training methods.[1] However, camps also often require athletes to travel and sleep in unfamiliar environments, which may negatively affect sleep.[2,3] Sleep is widely considered a major contributor to athletic recovery[4] and performance,[5] with evidence indicating that sleep disruption negatively affects psychological and physiological performance measures.[6,7] Investigating the influence of changing the training environment on the sleep of elite AF players may aid in the design of training interventions that minimize the potentially negative influences of poor sleep and recovery to maximize training adaptations.

Training camps often combine changes in training environment with additional environmental stressors such as altitude or heat.[8,9] In isolation, altitude and heat exposure have been shown to negatively influence objective measures of sleep quantity and quality.[10,11] However, the impact of changing training environment alone, such as during a training camp, on sleep quantity and quality is unknown. Changes in sleep location may disrupt sleep,[2] with ~28% of athletes from a German sample reporting unusual surroundings as disruptive to sleep on the night before competition.[3] AF players also regularly travel interstate for preseason camps and during the in-season period, resulting in reduced sleep durations when away from home on the night pregame.[12]

Another important consideration is the effect of increased training load on sleep. For example, Killer et al[13] reported disturbed sleep and mood state during a period of intensified training, while Kölling et al[14] reported less restful sleep in the first half of a 4-week training camp in elite youth rowers. Indeed, poor or disrupted sleep may negatively influence training adaptations and the efficacy of training interventions, an unfavorable situation for elite team-sport athletes. Given this, further research is required to uncover the influence of changing training and sleep location on quality and quantity of sleep during periods of high training load.

The aim of the current study was to assess the effects of a change in physical sleeping environment on objective and subjective measures of sleep quality and quantity in elite AF players during a training camp in the preseason training phase.

Methods

Nineteen professional AF players from the same Australian Football League club (age 22.1 ± 3.5 y, height 185.4 ± 7.6 cm, mass 83.3 ± 7.8 kg) participated in this study. Participants were free from musculoskeletal injury and illness before participating in the study, and no injuries occurred during the data-collection periods. Full informed consent was provided by participants, and the research was approved by the university human research ethics committee and followed guidelines from the Declaration of Helsinki.

Sleep was assessed using wristwatch activity devices across an 8-day home and 8-day camp period. Actiwatch 2 (Actiwatch 2,

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Philips Respironics, Murrysville, PA, USA) and ActiGraph wGTX3 (ActiGraph wGTX3 monitor, ActiGraph, Pensacola, FL, USA) devices were used interchangeably during the home period, and, as such, cross-device validity was assessed before measurement. Six participants from a separate sample both wore wrist-worn activity devices and completed subjective sleep diaries on each morning for 3 consecutive nights in the home environment. Concurrent validity of the devices was assessed using Pearson correlation coefficients and typical error of estimate (TEE).[15] Sleep efficiency (SE, $r = .92$, TEE = 2.1%), time in bed (TIB, $r = .97$, TEE = 2.9%), total sleep time (TST, $r = .91$, TEE = 6.2%), and wake after sleep onset (WASO, $r = .96$, TEE = 21.1%) demonstrated strong validity between devices. As such, these 4 objective measures were deemed suitable to determine sleep quantity (TIB, TST) and sleep quality (SE, WASO) for the current study. Sleep-onset latency was deemed not reliable ($r = .41$, TEE = 57%) and so was not used in the study.

During both the home and camp periods, participants completed 5 ± 1 training days and 3 ± 1 recovery days. Both the home (December 2014, n = 8, Actiwatch 2, or January 2015, n = 11, ActiGraph wGTX3) and camp (January to February 2015, n = 19, ActiGraph wGTX3) periods took place during the preseason training phase and followed similar time-of-day training schedules. Home training occurred at the club's training facility in Melbourne, Australia (latitude 37°s, longitude 144°E), with participants sleeping in their normal home-based sleeping environment (in their own beds); they were not grouped or paired with other participants. The camp occurred at the Sunshine Coast, Australia (latitude 27°s, longitude 153°E), with participants sleeping in a 4-star hotel with access to air conditioning and comfortable bedding. On camp, participants chose whom to room with and were subsequently placed in a shared living environment with another participant. Training time was similar between home and camp (commencing between 9:30 and 10:00 AM), and participants did not experience competition, transmeridian travel, or a change in altitude, leaving location as the difference between home and camp environments. Daytime training environmental conditions for home (23.6°C ± 4.4°C, 66.5% ± 3.2% RH) and camp (29.0°C ± 1.8°C, 67.8% ± 6.5% RH) were retrospectively obtained via Australia's Bureau of Meteorology (www.bom.gov.au). Overnight room temperature at both locations ranged from 18°C to 22°C and was kept constant through the use of air conditioning at the discretion of participants.

Subjective sleep diaries to collect bedtime, wake time, and a subjective sleep-quality rating were used daily. Based on recommendations by Hooper and Mackinnon,[16] a custom-designed daily subjective wellness questionnaire, where athletes subjectively rated measures of fatigue, stress, muscle soreness, sleep quality, and mood on a 1-to-10 scale (1 being worst possible, 10 being best possible), was completed each morning before daily training activities. Training sessions were designed and implemented by coaching staff, with training load assessed using the session rating of perceived exertion (s-RPE) (Borg CR-10 scale)—a valid measure of training load in team sports.[17] Participants undertook prescribed posttraining team-based recovery sessions on planned recovery days, consisting of cold-water immersion, stretching, self-myofascial release, and hydrotherapy.

Repeated-measures ANOVA were used to analyze differences in objective sleep measures (TIB, TST, SE [TST ÷ TIB × 100], and WASO), s-RPE training load, temperature, relative humidity and subjective wellness between home and camp. One-way ANOVA was used to screen sleep-onset and wake times for differences in sleep behavior. The summation of fatigue, muscle-soreness, stress, mood, and sleep-quality scores from the subjective wellness questionnaire represented total wellness, which has previously been used to track the response to competition in team-sport athletes.[18] Statistical significance was set to $P = .006$ via Bonferroni correction. Pearson correlation coefficients, controlling for repeated observations on players, were used to assess the relationship between sleep characteristics and s-RPE training load, relative camp sleep characteristics (expressed as absolute change from individual environment mean), and absolute change in s-RPE training load from the individual environment mean. Cohen effect sizes (d) were calculated using player mean values between home and camp (<0.2 = very small effect, 0.2–0.5 = small effect, 0.5–0.8 = moderate effect, >0.8 = large effect).

Results

Descriptive statistics for sleep characteristics and training-related parameters between home and camp are compared in Table 1. One-way ANOVA showed that sleep-onset ($P < .001$) and wake times ($P = .003$) were significantly earlier on camp than home (Table 1). Repeated-measures ANOVA revealed that TIB ($P < .001$) and WASO ($P < .001$) were significantly increased on camp compared with home. However, there was no significant change in TST ($P = .846$), resulting in a significant decrease in SE ($P < .001$) on camp. Repeated-measures ANOVA also showed that mean daily s-RPE training load ($P = .398$) and total player wellness ($P = .023$) were not significantly different on camp compared with home. Although training temperature ($P < .001$) during camp was significantly hotter than home, relative humidity ($P = .277$) did not differ (Table 1). Despite the observed changes in objective sleep measures, there was no difference between home and camp for subjective ratings of sleep quality (Table 1).

Effect-size calculations between grouped player mean values revealed a moderate effect of change in physical sleep environment on sleep onset ($d = –0.47$); a small effect on wake times ($d = –0.36$); large effects on TIB ($d = 1.21$), SE ($d = –0.93$), and WASO ($d = 0.87$); and only a very small effect on TST ($d = –0.07$). Effect-size calculations also revealed a very small effect of a change in training environment on s-RPE training load ($d = 0.10$), a small effect on total player wellness ($d = –0.36$), a large effect on daytime training temperatures ($d = 1.63$), and only a very small effect on daytime training humidity ($d = 0.13$).

When analyzing individualized responses to a change in physical sleep environment, relative changes in TIB ($r = –.75$, $P < .001$) and WASO ($r = –.72$, $P < .001$) on camp had a strong negative correlation with absolute home values (Figure 1). When accounting for daily variations of within-player training load, changes in s-RPE training load had weak correlations with changes in TST in the home environment for both increased ($r = .051$, $P = .74$) and decreased s-RPE training load ($r = –.033$, $P = .74$). Comparatively, in the camp environment, changes in s-RPE training load displayed a moderate negative correlation with changes in TST for increased load ($r = –.367$, $P = .010$) and a moderate positive correlation for decreased load ($r = .319$, $P = .003$) (Figure 2).

Discussion

The aim of the current study was to determine the effect of changing physical sleep environment on objective sleep quality and quantity in elite AF players during the preseason training period. This study found that during camp players spent longer in bed, as a result of earlier sleep-onset times that were not matched by equal alterations

Table 1 Difference in Sleep and Training Characteristics Between Home and Camp in Elite Australian Football Players, Mean ± SD

	Home	Camp	Effect size (95%CI)	*P*
Sleep-specific measures				
bedtime (h:min)	22:25 ± 02:02	21:40 ± 00:46	–0.47 (–0.71, –0.24)	<.001*
wake time (h:min)	06:51 ± 01:03	06:31 ± 00:44	–0.36 (–0.59, –0.13)	.003*
time in bed (min)	496.5 ± 31.5	530.6 ± 24.3	1.21 (0.50, 1.88)	<.001*
total sleep time (min)	419.4 ± 25.6	417.0 ± 37.5	–0.07 (–0.71, 0.56)	.846
sleep efficiency (%)	84.7 ± 6.5	78.7 ± 6.5	–0.93 (–1.57, –0.24)	<.001*
wake after sleep onset (min)	68.9 ± 29.9	96.5 ± 33.7	0.87 (0.19, 1.51)	<.001*
subjective sleep quality (AU)	3.50 ± 0.43	3.46 ± 0.52	–0.08 (–0.72, 0.55)	.946
Training-specific measures				
session rating of perceived exertion (AU)	396 ± 394	440 ± 502	0.10 (–0.13, 0.32)	.398
temperature (°C)	23.6 ± 4.4	29.0 ± 1.8	1.63 (1.36, 1.88)	<.001*
relative humidity (%)	66.5 ± 13.2	67.8 ± 6.5	0.13 (0.10, 0.35)	.277
total wellness (AU)	39.7 ± 2.9	38.8 ± 2.2	–0.36 (–0.65, –0.07)	.023

Abbreviations: AU, arbitrary units; CI, confidence intervals.
*Significant difference ($P < .006$).

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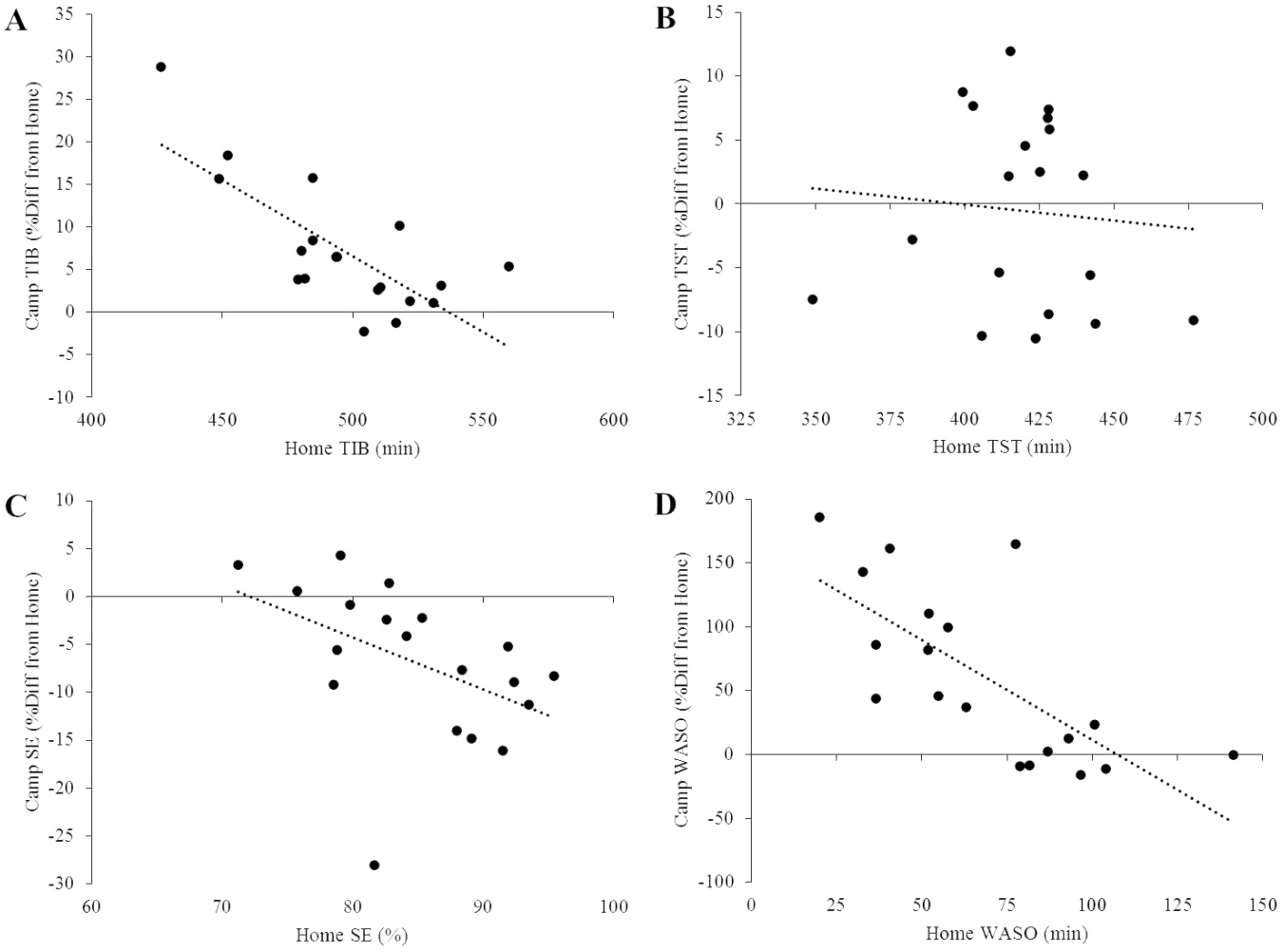

Figure 1 — Sleep characteristics on camp expressed as relative individual changes from mean home (A) time in bed, (B) total sleep time, (C) sleep efficiency, and (D) wake after sleep onset.

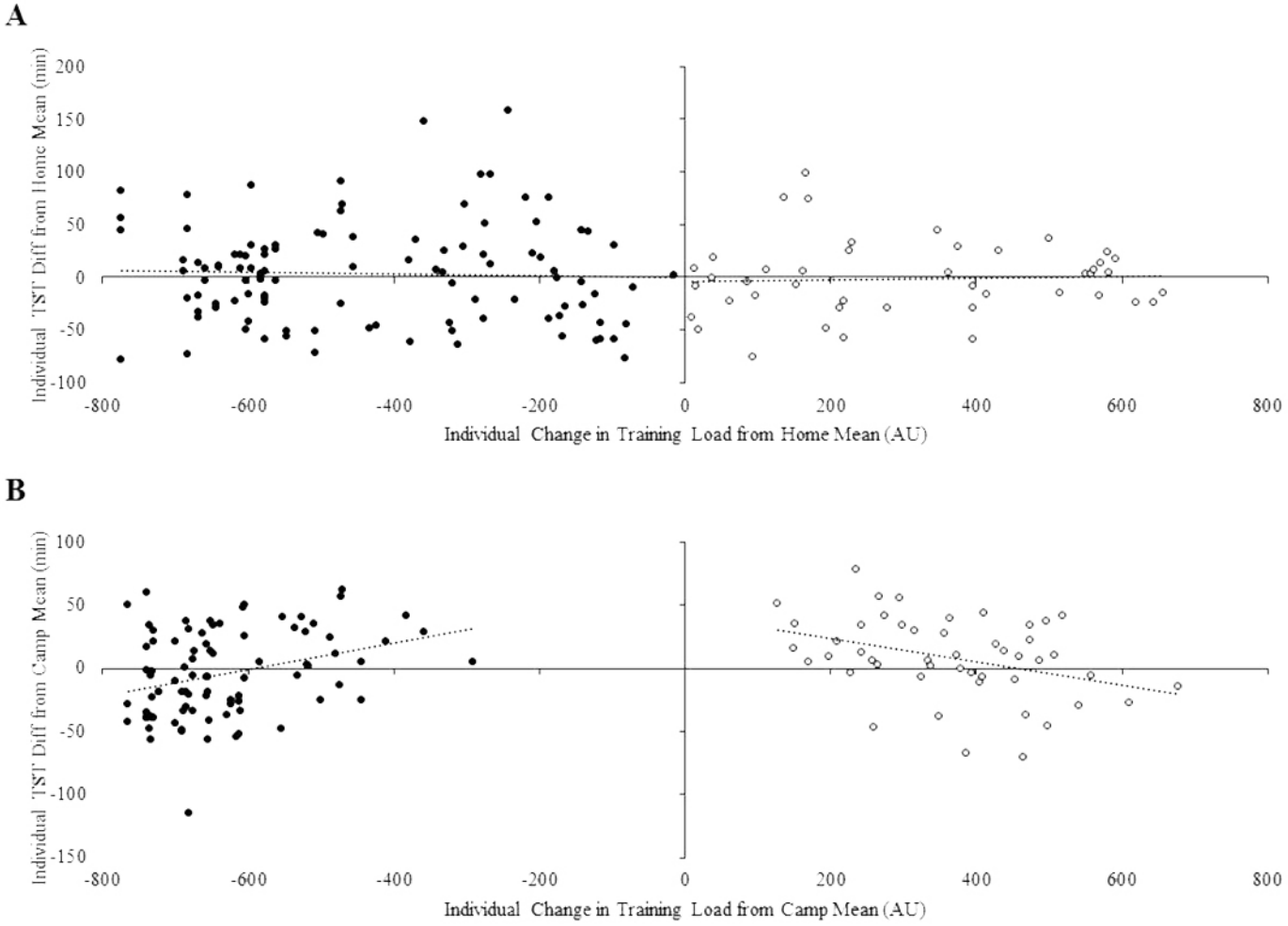

Figure 2 — Variations in total sleep time from the environment mean in response to increased (open circles) and decreased (solid circles) daily training load from the environment mean (A) at home and (B) on camp.

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in wake times, and exhibited more WASO, resulting in lower SE than at home. Furthermore, players who spent less TIB at home increased TIB to a greater extent on camp than those who spent longer in bed at home. While these results are independent of mean daily training load, when changes in the individual daily training load are accounted for, alterations in daily training load on camp resulted in greater changes in TST than at home. Together, these results have relevance for practitioners in highlighting the need for individualized sleep strategies to overcome the negative effects of changes in sleep environment.

This is the first study to observe a negative effect of changing sleep environment on sleep quality during periods of similar training loads in team-sport athletes. We found that athletes with higher sleep quality at home were worse sleepers at camp, whereas those who slept poorer at home slept similarly on camp. Sleep quality in athletes can be affected by numerous factors such as travel,[19] competition,[20] and training load.[21] Furthermore, Lastella et al[22] have shown that increased daily competition load in cyclists significantly reduces TIB and TST, without changes in subjective or objective sleep quality. The current study showed that a change from home to camp, where participants stayed in a hotel and roomed with a teammate, negatively affected sleep quality, despite no difference in mean training load. While poor sleep can result from changes in training load,[13,21] these changes have predominantly been assessed on a global scale, without consideration for an individual's daily variation in load and sleep. Indeed, when accounting for variations in daily training load, this study found that when there is a change in sleep location, TST responds in a different manner. For example, on camp, as the change in daily training load varied further from the mean training load for each individual, there was a reduction in TST (Figure 2). Although these results appear consistent with previous research,[21,23] the reason for this is unclear, and so further research is required to determine the longer-term effect on training adaptation.

Studies investigating changes in physical sleeping environment have focused on situations where participants travel across multiple time zones, creating a circadian dysrhythmia and reduced sleep quality.[24] Although the current study involved travel, it was in a northerly direction during the day, without crossing time zones and without affecting participants' sleep opportunities. Furthermore, camp situations are often accompanied by altered daily schedules, as a result of commitments outside of training, which may affect sleep behavior. Although sleep-onset and wake times were earlier on camp in the current study, this did not appear to be a result of altered daily schedules as there was no change between home and camp. However, it should be noted that due to the change in latitude, there is a small change in daylight savings, with earlier sunrise and

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sunset on camp. Hence, it should not be discounted that the earlier sunset influenced the choice of bedtime each night.

Previous work involving changes in physical sleeping environment has examined tournament-type competition[20] or regular/weekly competition[12] and compared sleep during periods of competition in a different physical environment with sleep during periods of training at home.[22,25] Lastella et al[25] found that ~68% of athletes report poorer sleep quality on nights before competition and that this is often related to feelings of anxiety, fatigue, mood, tension, and vigor. Although no competition occurred in the camp environment, athletes in the current study spent significantly longer in bed, without the benefits of increased sleep time. Indeed, WASO was ~30 minutes higher on camp and therefore may explain the difference in TIB between home and camp (~34 min). However, due to the very high TEE (57%) between activity devices for sleep-onset-latency measurements, this study is limited in its application of this measure. In this regard, this should be seen as a limitation of the current results. Nevertheless, future work should aim to identify the behavioral changes that are associated with changes in sleep behavior.

The consideration of non-training-related factors relevant to the physical environment is also important. Ambient environmental temperature can negatively affect sleep, with exposure to heat before and during the sleep period reducing TST and increasing nighttime awakenings.[26] The current study showed significantly higher daytime temperatures on camp than at home, corresponding with increased WASO and reduced SE, but no change in TST. However, the temperature difference between home and camp in the current study was much less than in previous literature and did not reach temperatures commonly used in studies of heat exposure and sleep.[10,26] Furthermore, nighttime temperatures in this study were athlete-controlled (18–22°C) through the use of air conditioning, thus in keeping with recent sleep-hygiene recommendations for athletic populations.[27] Although it cannot be entirely ruled out that the elevated daytime temperature on camp influenced sleep in the current study, this does not explain the increased TIB and the lack of overnight heat being reported as a reason for disrupted sleep from the subjective sleep diaries. In addition, it is difficult to say whether air conditioning had any impact on WASO. Thus, future research should examine the impact of both air conditioning and ambient overnight temperature on sleep. It is important to note that it is a possibility that the change in sleep behavior may be driven by a conscious process of respecting their teammates' space and noise, as all participants in the current study shared rooms with another participant during the camp period only. The degree and variation in the consideration of fellow participants may also vary depending on the individual relationships and habits of room-sharing participants and therefore should be considered in the interpretation of findings. Despite the need for individualized approaches to promote optimal sleep opportunities, the individualized nature of responses to changes in the physical sleep environment requires greater investigation. Together, this may have implications for the provision of sleep interventions and sleep-hygiene recommendations.

In the current study we observed that those with higher home SE experienced a greater decline in SE during camp. Hence, it is important that individually tailored sleep interventions are implemented to maximize TST. For instance, if participants in the current study maintained home SE of 84.7% while on camp, TST as a result of increased TIB would have been extended by 29 minutes. Indeed, sleep extension in basketball players improved sprint time, free-throw accuracy, reaction time, and ratings of physical and mental well-being.[28] Theoretically, maximizing TST and therefore recovery during a period of high training stress may accentuate training adaptations and aid in reducing the risk of injury.[29] However, given the lack of evidence on sleep extension in athletes, future research should aim to uncover the smallest worthwhile increase in sleep needed to improve training and/or competition performance.

Conclusion

A change in physical sleep environment, without external influences such as circadian phase shifting, altered training schedules, or increased total training load, reduces the quality of sleep and affects the attainment of increased duration of sleep, despite spending longer periods of time in bed. This may have further flow-on effects on recovery and the adaptive training response during periods where physiological adaptations are crucial to performance. Furthermore, the individual variation in response to a change in environment stresses the importance of assessing sleep on a case-by-case basis, especially if assessment leads to the provision of interventions designed to improve the sleep of athletes during time spent in unfamiliar physical sleeping environments.

Practical Implications

- Efforts should be made to enhance sleep duration when changing physical sleep environments to maintain sleep quality.
- Understanding that individuals respond differently to changes in the physical sleeping environment may help improve sleep quality and quantity of traveling athletes or athletes who are required to regularly sleep in differing physical environments.
- Fluctuations in individual daily training loads when on camp may make athletes more susceptible to altered sleep responses, highlighting the need for individualized interventions.

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References

1. Buchheit M, Racinais S, Bilsborough JC, et al. Monitoring fitness, fatigue and running performance during a pre-season training camp in elite football players. *J Sci Med Sport*. 2013;16(6):550–555. PubMed doi:10.1016/j.jsams.2012.12.003
2. Bishop D. The effects of travel on team performance in the Australian national netball competition. *J Sci Med Sport*. 2004;7(1):118–122. PubMed doi:10.1016/S1440-2440(04)80050-1
3. Erlacher D, Ehrlenspiel F, Adegbesan OA, et al. Sleep habits in German athletes before important competitions or games. *J Sports Sci*. 2011;29(8):859–866. PubMed doi:10.1080/02640414.2011.565782
4. Lastella M, Roach GD, Halson SL, Martin DT, West NP, Sargent C. The impact of a simulated grand tour on sleep, mood, and well-being of competitive cyclists. *J Sports Med Phys Fit*. 2014;55(12): 1555–1564.
5. Samuels C. Sleep, recovery, and performance: the new frontier in high-performance athletics. *Neurol Clin*. 2008;26(1):169–180, ix–x. PubMed doi:10.1016/j.ncl.2007.11.012
6. Belenky G, Wesensten NJ, Thorne DR, et al. Patterns of performance degradation and restoration during sleep restriction and subsequent

recovery: a sleep dose-response study. *J Sleep Res*. 2003;12(1):1–12. PubMed doi:10.1046/j.1365-2869.2003.00337.x
7. Mougin F, Simon-Rigaud ML, Davenne D, et al. Effects of sleep disturbances on subsequent physical performance. *Eur J Appl Physiol Occup Physiol*. 1991;63(2):77–82. PubMed doi:10.1007/BF00235173
8. Buchheit M, Racinais S, Bilsborough J, et al. Adding heat to the live-high train-low altitude model: a practical insight from professional football. *Br J Sports Med*. 2013;47(Suppl 1):i59–i69. PubMed doi:10.1136/bjsports-2013-092559
9. Sargent C, Schmidt WF, Aughey RJ, et al. The impact of altitude on the sleep of young elite soccer players (ISA3600). *Br J Sports Med*. 2013;47(Suppl 1):i86–i92. PubMed doi:10.1136/bjsports-2013-092829
10. Okamoto-Mizuno K, Tsuzuki K, Mizuno K, et al. Effects of partial humid heat exposure during different segments of sleep on human sleep stages and body temperature. *Physiol Behav*. 2005;83(5):759–765. PubMed doi:10.1016/j.physbeh.2004.09.009
11. Roach GD, Schmidt WF, Aughey RJ, et al. The sleep of elite athletes at sea level and high altitude: a comparison of sea-level natives and high-altitude natives (ISA3600). *Br J Sports Med*. 2013;47(Suppl 1):i114–i120. PubMed doi:10.1136/bjsports-2013-092843
12. Richmond L, Dawson B, Hillman DR, et al. The effect of interstate travel on sleep patterns of elite Australian Rules footballers. *J Sci Med Sport*. 2004;7(2):186–196. PubMed doi:10.1016/S1440-2440(04)80008-2
13. Killer SC, Svendsen IS, Jeukendrup AE, Gleeson M. Evidence of disturbed sleep and mood state in well-trained athletes during short-term intensified training with and without a high carbohydrate nutritional intervention [published online ahead of print September 25, 2015]. *J Sports Sci*. doi:10.1080/02640414.2015.1085589. PubMed
14. Kölling S, Steinacker JM, Endler S, Ferrauti A, Meyer T, Kellmann M. The longer the better: sleep–wake patterns during preparation of the World Rowing Junior Championships. *Chronobiol Int*. 2016;33(1):73–84. doi:10.3109/07420528.2015.1118384.
15. Hopkins WG. Measures of reliability in sports medicine and science. *Sports Med*. 2000;30(1):1–15. PubMed doi:10.2165/00007256-200030010-00001
16. Hooper SL, Mackinnon LT. Monitoring overtraining in athletes: recommendations. *Sports Med*. 1995;20(5):321–327. PubMed doi:10.2165/00007256-199520050-00003
17. Scott TJ, Black CR, Quinn J, et al. Validity and reliability of the session-RPE method for quantifying training in Australian football: a comparison of the CR10 and CR100 scales. *J Strength Cond Res*. 2013;27(1):270–276. PubMed doi:10.1519/JSC.0b013e3182541d2e
18. McLean BD, Coutts AJ, Kelly V, et al. Neuromuscular, endocrine, and perceptual fatigue responses during different length between-match microcycles in professional rugby league players. *Int J Sports Physiol Perform*. 2010;5(3):367–383. PubMed doi:10.1123/ijspp.5.3.367
19. Reilly T, Waterhouse J, Edwards B. Jet lag and air travel: implications for performance. *Clin Sports Med*. 2005;24(2):367–380, xii. PubMed doi:10.1016/j.csm.2004.12.004
20. Fullagar HH, Duffield R, Skorski S, et al. Sleep, travel, and recovery responses of national footballers during and after long-haul international air travel. *Int J Sports Physiol Perform*. 2016;11(1):86–95. https://doi.org/10.1123/ijspp.2015-0012
21. Taylor SR, Rogers GG, Driver HS. Effects of training volume on sleep, psychological, and selected physiological profiles of elite female swimmers. *Med Sci Sports Exerc*. 1997;29(5):688–693. PubMed doi:10.1097/00005768-199705000-00016
22. Lastella M, Roach GD, Halson SL, et al. Sleep/wake behaviour of endurance cyclists before and during competition. *J Sports Sci*. 2015;33(3):293–299. PubMed doi:10.1080/02640414.2014.942690
23. Hausswirth C, Louis J, Aubry A, et al. Evidence of disturbed sleep and increased illness in overreached endurance athletes. *Med Sci Sports Exerc*. 2014;46(5):1036–1045. PubMed doi:10.1249/MSS.0000000000000177
24. Lastella M, Roach GD, Halson SL, et al. The effects of transmeridian travel and altitude on sleep: preparation for football competition. *J Sports Sci Med*. 2014;13(3):718–720. PubMed
25. Lastella M, Lovell GP, Sargent C. Athletes' precompetitive sleep behaviour and its relationship with subsequent precompetitive mood and performance. *Eur J Sport Sci*. 2014;14(Suppl 1):S123–S130. PubMed doi:10.1080/17461391.2012.660505
26. Libert JP, Di Nisi J, Fukuda H, et al. Effect of continuous heat exposure on sleep stages in humans. *Sleep*. 1988;11(2):195–209. PubMed
27. Nédélec M, Halson S, Delecroix B, et al. Sleep hygiene and recovery strategies in elite soccer players. *Sports Med*. 2015;45(11):1547–1559. PubMed doi:10.1007/s40279-015-0377-9
28. Mah CD, Mah KE, Kezirian EJ, et al. The effects of sleep extension on the athletic performance of collegiate basketball players. *Sleep*. 2011;34(7):943–950. PubMed doi:10.5665/sleep.1132
29. Milewski MD, Skaggs DL, Bishop GA, et al. Chronic lack of sleep is associated with increased sports injuries in adolescent athletes. *J Pediatr Orthop*. 2014;34(2):129–133. PubMed doi:10.1097/BPO.0000000000000151

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